EXP. No: 02
DATE : 15/07/2026

LEARNING AND ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTER.

AIM :-

To learn IP addressing & manually assign an IP address to a computer. To verify the assigned IP address using Linux networking commands.

THEORY :-

An IP address is a unique numerical address assigned to every device connected to a network. It enables communication between devices over a network. An IPv4 address consists of four octets seperated by dots (eg: 192.168.1.10)

An IP address can be assigned either manually (static IP) or automatically (dynamic IP) using DHCP. A manually assigned IP address remains unless changed by the user. Along with the IP address, parameters such as the subnet mask, default Gateway, & DNS server are also configured for proper network communication.

In Linux, the ip command is used to display & manage IP address, network interfaces, routing tables, and neighbouring devices. It is one of the most commonly used commands for network configuration & troubleshooting.

COMMANDS AND SAMPLE OUTPUTS :-

1) Display all IP address :-

Command :

ip address show

Sample output :

2 : ens160 :

inet 172.16.211.128/24

inet6 fe80::20c:29ff::5e75:9927/64.

Observation :

Displays the IPV4 & IPV6 addresses of all networks.

2) Display a specific network interface :-

Command :

ip addr show dev ens160.

Sample output :

2 : ens160 :

inet 172.16.211.128/24

3) Display Routing Table :-

Command :

ip route show

Sample output :

default via 172.16.211.2 dev ens160

172.16.211.0/24 dev ens160

Observation :

Displays the routing table & the default gateway.

4) Display Neighbor (ARP) table :-

Command :

ip neigh show

Sample output :

172.16.211.1 REACHABLE

172.16.211.254 STABLE

Observation :

Displays the neighbouring devices & their MAC address.

RESULT :-

The IP address was manually assigned & verified successfully using Linux networking commands.

15/2/26